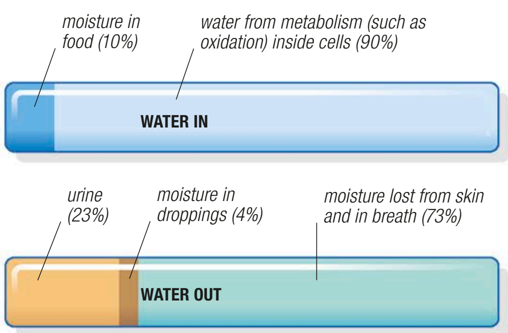
Life in desert

In a habitat where moisture is scarce, obtaining and conserving water are every animal's top priorities. Desert animals practice a tight "water economy," which means collecting water wherever they can, and minimizing water loss wherever possible. However, being economical with water is not in itself enough to guarantee survival: desert species have had to evolve various other adaptations to enable them to cope with a wide range of temperatures and the ever-present threat of food shortage. As a result, these animals are able to live in some of the driest places on earth.

Conserving water

Most deserts have a scattering of oases, where animals gather to drink. Some species need to drink daily, which restricts how far they can roam from an oasis. Others can survive on their on-board reserves for days or even weeks, depending on the temperature. A remarkable feature of desert life is that some animals can manage without drinking at all. Instead, they get all their water from their food. Some extract it from the moisture contained in food; but most use the food to manufacture metabolic water, which is created by chemical reactions when the energy in food is released. Seed-eating rodents are expert at this: although their food looks dry, they are able to metabolize all the water they need.



WATER BALANCE
This diagram illustrates how a kangaroo rat survives entirely on the water in its food. The water taken in has to balance that which is lost to prevent the animal from becoming dehydrated.

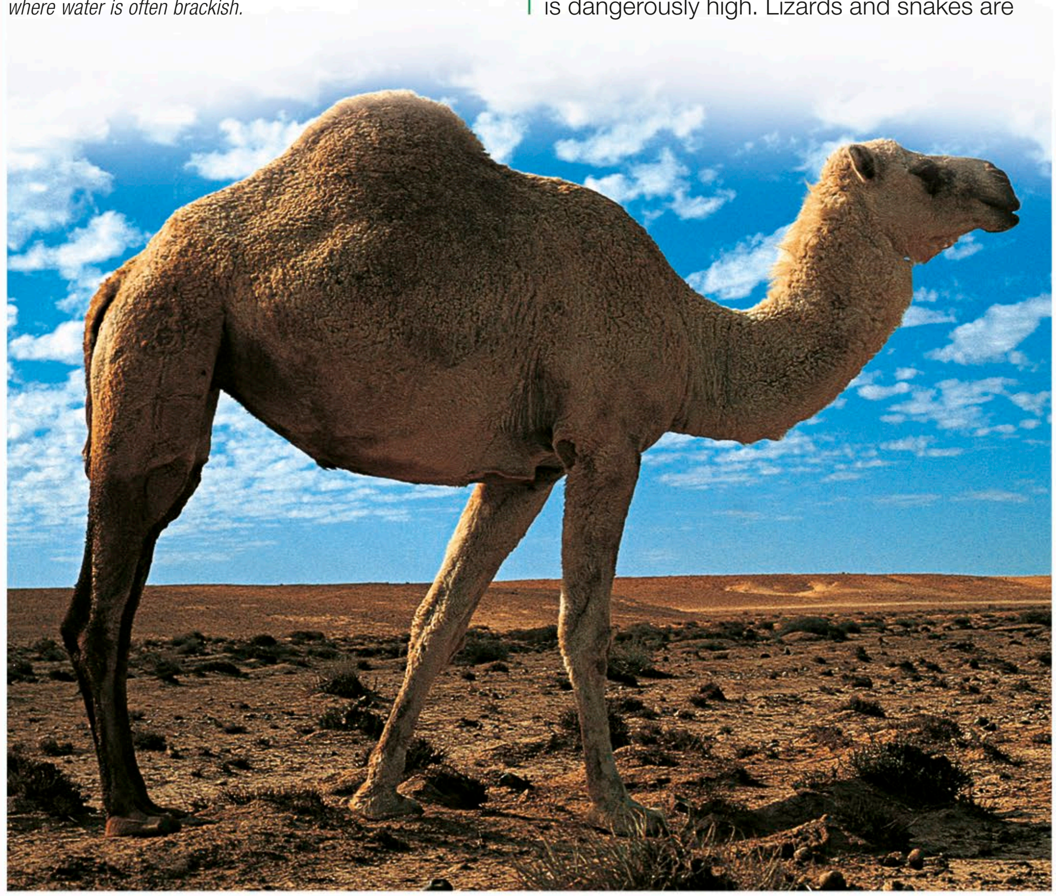
WATER-STORING FROG
The Australian water-holding frog stores water in its bladder and beneath its skin. To prevent this water from drying out, the frog then seals itself in a semipermeable cocoon underground.

For drinkers and nondrinkers alike, water has to be eked out to make sure that it lasts. Compared with animals from other habitats, desert species lose very little moisture in their urine and droppings, and only a small amount is released from their skin and in their breath. Desert species are also good at withstanding dehydration. The dromedary, or one-humped camel, can lose nearly one half of its body water and survive. For humans, losing just a fifth can be fatal.

Storing food

To enable them to cope with erratic food supplies, many animals keep their own food reserves. Some do this by hiding food away. The North American kangaroo rat, for example, constructs underground granaries that contain up to 11 lb (5 kg) of seeds. But for predators, and for animals that browse on shrubs, creating such larders is not possible. Their food is difficult to collect and to transport, and even if it could be hoarded it would be unlikely to remain usable for more than just a few days. The answer is to store food inside the body. The classic example of this is the camel, which stores surplus food, in the form of fat, in its hump. Several other species, such as the Gila monster and fat-tailed dunnart, store food in their tails.

RAPID REFILL
After going without water for several days, a camel can drink over 11 gallons (50 liters) in just a few minutes. It also metabolizes water from surplus food, laid down as fat in its hump. Its salt tolerance is high, which is useful in a habitat where water is often brackish.



Coping with heat and cold

In desert, the temperature rarely stays steady for more than a few hours, and it can reach extremes of both heat and cold very quickly. Humans lose excess heat by sweating, but at very high temperatures, this cooling system can use as much as 35 fl. oz (1 liter) of water an hour—far more than any desert animal could afford.

Desert animals tackle the heat problem in 2 ways: by reducing the heat they absorb, and by increasing the heat they give out. Light-colored skin or fur reflects some of the sun's rays, minimizing heat absorption; but a much more effective method—used by many desert animals—is to avoid the most intense heat by being nocturnal, spending the day sheltering underground. Burrows do not have to be very deep to make a difference: while the desert surface may be too hot to touch, the ground just a few inches below it will be relatively cool.

Getting rid of excess heat is more difficult, particularly when an animal's body temperature is dangerously high. Lizards and snakes are